SAM 3 and Esri's Mask R-CNN

May 2026 Nearmap imagery, Lewisville, Texas. Historical inventory agreement, not reviewed 2026 accuracy.

Meta's SAM 3 and Esri's Building Footprint Extraction - USA model ran on the same city-wide aerial imagery at 10.16 cm per pixel. This report preserves the recorded results, frozen settings and eight screenshots.

Reference provenance: 38,924 source inventory features, dated primarily 2015 or earlier, with 33 undated. The city-wide evaluation contains 38,882 outlines. These are separate source and evaluated counts. Collection dates do not establish construction or demolition dates.

Imagery: Nearmap, May 2026 · 29 September 2026 · interpretation and provenance revised 15 October 2026

0.95 / 0.91 / 0.87

Held-out test area

Historical F1 for the city's Esri run, local Esri run and SAM 3, respectively, reported for buildings of at least 20 m^2. See the scoring audit for matching-policy evidence.

0.83 vs 0.74

City-wide ≥20 m^2

The city's Esri run leads SAM 3 in inventory agreement. Recall is 0.85 and 0.82; precision is 0.82 and 0.68.

91%

Model agreement

Share of the city's Esri outlines with a SAM 3 partner. Matching models can share errors; agreement does not supply independent labels.

4,923

Review candidates

SAM 3 outlines without an inventory match that also match Esri. Inventory omissions, geometry conventions, change and shared errors remain possible.

7,599

Small structures missed

Inventory outlines under 20 m^2 missed by both the city's Esri run and SAM 3, out of 7,867. SAM 3 found 267; the city's Esri run found 2.

Review pending

No accuracy claim

No completed blind human review or review receipt is recorded. The local review packages do not establish that the labels were checked against the imagery.

An unmatched or missed outline is not dated evidence of construction, change or demolition.

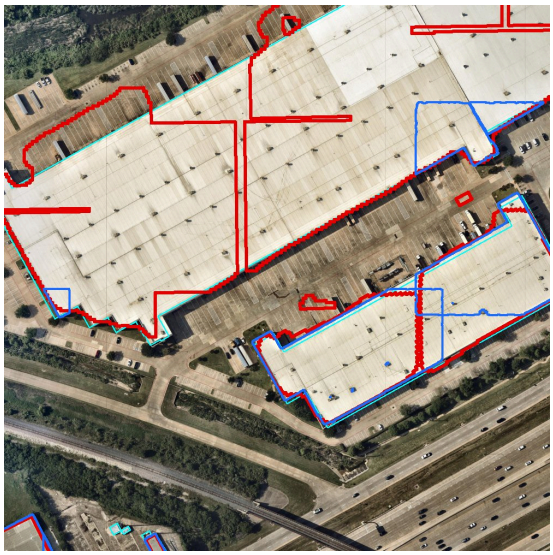
Source details: reference provenance and scoring audit.

Screenshots: model agreement and differences

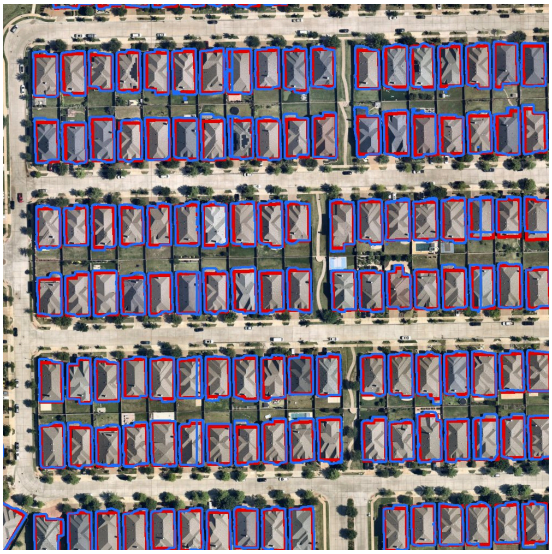
Red: the city's Esri run. Blue: SAM 3, "building", 1536 px, confidence threshold 0.7. Cyan: historical inventory.



Typical residential area. 400 m square (33.0555° N, 96.9029° W). Both models outline almost every visible house. A few blue lines cross roofs where SAM 3 drew overlapping outlines.



Distribution centre. 400 m square (33.0558° N, 96.9157° W). Esri splits the largest roof at its chip edges. SAM 3 finds a corner and splits the smaller building where tiles meet.



Inventory-unmatched residential area. 300 m square (33.0255° N, 96.8972° W). Both models outline visible houses without inventory counterparts. This square has 125 shared outlines without an inventory match. No construction date is established.



SAM 3's additional outlines. 300 m square (32.9923° N, 97.0167° W). A manufactured-home community beside a warehouse. The square has 57 SAM 3 outlines unmatched to both Esri and the inventory; they require review.

Eight existing screenshots: two typical areas and six 300 m squares selected by difference counts. They are outside the review AOIs. Held-out annotation must still be conducted blind to predictions and these candidate views. Observations refer to May 2026 imagery, not verified dates of change.

Screenshots: missed structures and tile edges

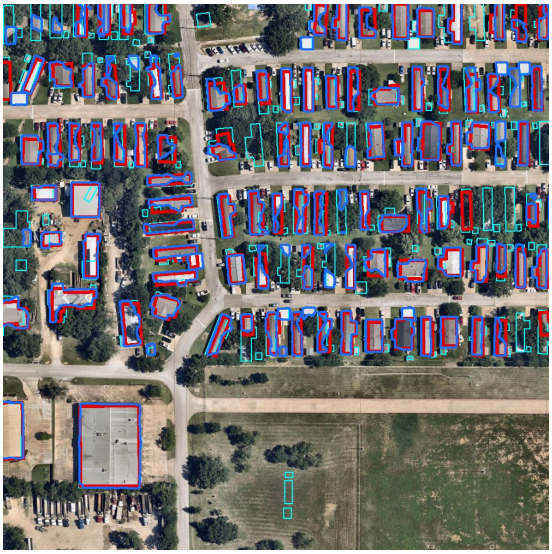
Red: the city's Esri run. Blue: SAM 3, "building", 1536 px, confidence threshold 0.7. Cyan: historical inventory.



Where Esri finds more. 300 m square (33.0107° N, 96.9842° W). At an apartment complex, Esri outlines long carport and garage rows that SAM 3 mostly misses. The square has 30 inventory outlines found only by Esri.



Where SAM 3 finds more. 300 m square (33.0764° N, 97.0341° W). SAM 3 outlines more back-yard garages and sheds. The square has 24 inventory outlines found only by SAM 3; many small cyan outlines remain unmatched.



Small structures and open ground. 300 m square (33.0324° N, 96.9869° W). The square has 61 inventory outlines under 20 m² missed by both. Some cyan outlines overlap apparently open ground. This observation does not establish demolition or its date.



Large buildings. 300 m square (33.0680° N, 97.0150° W). Both models find long industrial roofs. SAM 3 splits them where tiles meet, and Esri edges step at chip boundaries.

Eight existing screenshots: two typical areas and six 300 m squares selected by difference counts. They are outside the review AOIs. Held-out annotation must still be conducted blind to predictions and these candidate views. Observations refer to May 2026 imagery, not verified dates of change.

Imagery: Nearmap, May 2026.

Whole-city inventory agreement

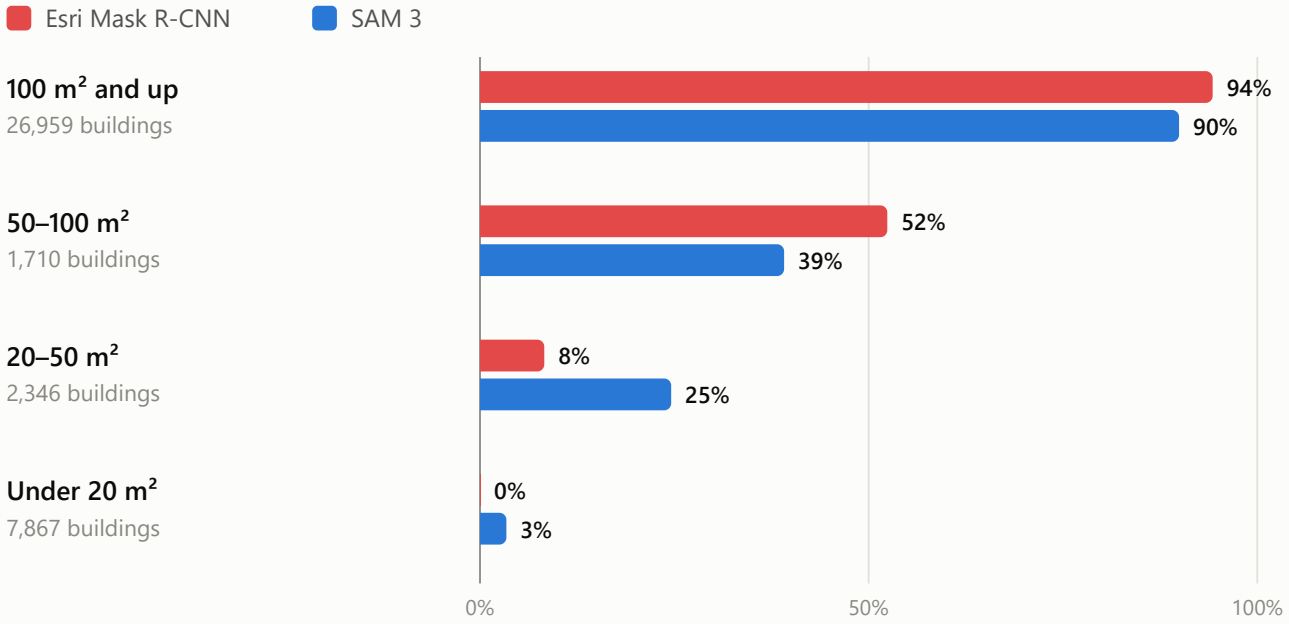
All three runs used shared cleanup and 38,882 evaluated inventory outlines. SAM 3 used 6,538 tiles; the local Esri run used 150 chunks.

Method	Outlines	TP	FP	FN	Precision	Recall	F1	F1 ≥20 m²	Median IoU
Esri, city run	32,608	26,502	6,106	12,380	0.81	0.68	0.74	0.83	0.77
SAM 3	37,906	25,758	12,148	13,124	0.68	0.66	0.67	0.74	0.80
Esri, local raw run	38,860	26,536	12,324	12,346	0.68	0.68	0.68	0.77	0.80

The ≥20 m² column retains the historical published scores; see the scoring audit for execution policy. Other columns use all sizes. Extra outlines need independent review for omissions, conventions, registration and model errors.

Both models find 9 in 10 large buildings, but few small ones

Recall by area: the share of 38,882 evaluated inventory outlines each model found



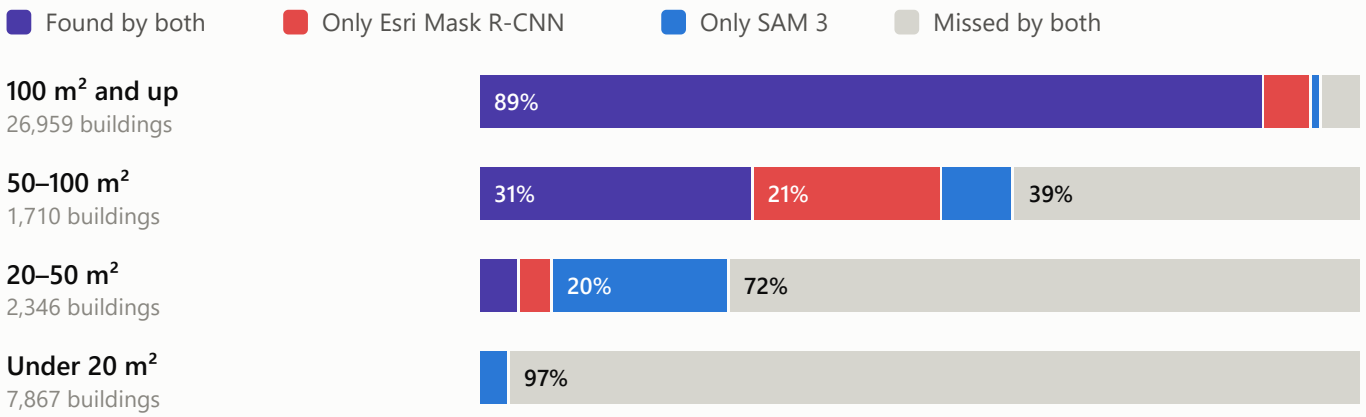
Primarily 2015 or earlier; 33 source features undated. Agreement, not reviewed 2026 accuracy.
Esri Mask R-CNN: the city's run of Esri's Building Footprint Extraction – USA model.

Area	Inventory	Found by Esri	Found by SAM 3
under 20 m²	7,867	2	267
20–50 m²	2,346	194	577
50–100 m²	1,710	896	669
100 m² and up	26,959	25,410	24,245

Outcomes and model-agreement candidates

Both models find most large buildings; neither finds most small ones

What happened to each evaluated inventory outline, by building area



Whole city, May 2026 imagery. Missed inventory outlines do not establish demolition.
Esri Mask R-CNN: the city's run of Esri's Building Footprint Extraction – USA model.

Area	Both	Only Esri	Only SAM 3	Neither
under 20 m²	1	1	266	7,599
20–50 m²	107	87	470	1,682
50–100 m²	530	366	139	675
100 m² and up	23,947	1,463	298	1,251

Both models found 24,585 inventory outlines; Esri alone found 1,917 and SAM 3 alone 1,173. Neither found 11,207, including 7,599 under 20 m². Adding local Esri, 10,463 were missed by all three, including 7,359 under 20 m².

Model-to-model agreement

29,567	91%	78%	0.81
matched pairs	of Esri outlines	of SAM 3 outlines	median IoU

Matching uses IoU ≥ 0.5. Of SAM 3's 12,148 inventory-unmatched outlines, 4,923 match Esri. These are review candidates. An older inventory may omit structures or use a different geometry convention; both models may also make the same error.

The compatibility name `candidate_new` does not mean that a candidate is new. Do not use this layer to select or correct blind held-out annotations.

Old Town: the untuned baseline

A 700 m square with 319 inventory outlines. Each method used default settings with the same cleanup and scoring. SAM 3 used 1024 px tiles and confidence threshold 0.5; local Esri used 0.5.

Method	TP	FP	FN	Precision	Recall	F1	Median IoU
SAM 3	142	287	177	0.33	0.45	0.38	0.74
Esri, city run	135	63	184	0.68	0.42	0.52	0.75
Esri, local raw run	158	381	161	0.29	0.50	0.37	0.74

Method	Recall ≥50 m²	Precision ≥50 m²	Recall <50 m²
SAM 3	0.63	0.39	0.10
Esri, city run	0.64	0.69	0.01
Esri, local raw run	0.71	0.36	0.10

Per building: 109 found by all three methods, 38 by two, 32 by one, and 140 by none. Of the 140 missed by all, 96 are under 50 m². These per-building counts are retained from the original report.

Interpretation boundary

Differences between Old Town and the test AOI do not establish when structures were built, altered or demolished. Density, object size, inventory completeness, geometry convention, occlusion and model behaviour remain possible explanations.

The baseline table uses all sizes. The tuning protocol requires filtering both sets to at least 20 m² before matching. Historical execution order must be checked separately.

Tuning in Old Town

The fixed objective was F1 at IoU ≥0.5 on buildings of at least 20 m². With no reviewed labels available, tuning used the historical inventory under Amendment 1. All 176 variants are retained in tuning_final.csv. Regularization did not change a score; shown off.



Method	Setting	Precision	Recall	F1	Median IoU	Recall <20 m²
Esri city run	as published (frozen)	0.69	0.55	0.61	0.75	0.00
SAM 3	building, 2048 px, ≥0.75	0.60	0.48	0.54	0.76	0.00
Esri local	confidence ≥0.9 (frozen)	0.56	0.50	0.53	0.76	0.03
SAM 3	building, 1536 px, ≥0.7 (frozen)	0.53	0.52	0.53	0.75	0.03
SAM 3	house, 1536 px, ≥0.7	0.60	0.42	0.50	0.75	0.03
SAM 3	roof, 2048 px, ≥0.75	0.54	0.46	0.49	0.75	0.01
SAM 3	house, 2048 px, ≥0.5	0.49	0.48	0.48	0.74	0.04
SAM 3	building, 1024 px, ≥0.75	0.48	0.47	0.48	0.74	0.01
SAM 3	roof, 1536 px, ≥0.7	0.41	0.50	0.45	0.75	0.03
SAM 3	house, 1024 px, ≥0.4	0.38	0.51	0.44	0.73	0.14
SAM 3	roof, 1024 px, ≥0.8	0.38	0.43	0.41	0.74	0.04

Confidence thresholds and the frozen rule

SAM 3 F1 by prompt, tile size and confidence threshold

Prompt	Tile size	≥0.3	≥0.4	≥0.5	≥0.6	≥0.7	≥0.75	≥0.8	≥0.85	≥0.9
building	1024 px	0.38	0.41	0.43	0.44	0.46	0.48	0.47	0.39	0.19
building	1536 px	0.43	0.46	0.46	0.49	0.53	0.52	0.51	0.47	0.28
building	2048 px	0.43	0.46	0.47	0.51	0.53	0.54	0.50	0.44	0.26
house	1024 px	0.43	0.44	0.43	0.43	0.42	0.42	0.41	0.36	0.17
house	1536 px	0.44	0.47	0.46	0.47	0.50	0.47	0.45	0.38	0.18
house	2048 px	0.44	0.47	0.48	0.48	0.45	0.44	0.41	0.36	0.17
roof	1024 px	0.29	0.31	0.34	0.35	0.36	0.37	0.41	0.40	0.29
roof	1536 px	0.34	0.37	0.39	0.41	0.45	0.45	0.45	0.42	0.33
roof	2048 px	0.36	0.40	0.42	0.44	0.47	0.49	0.46	0.39	0.26

Bigger tiles and higher confidence thresholds help up to about 0.75; above that, recall falls faster than precision rises. The best SAM 3 setting, 2048 px at 0.75 (F1 0.54), is within 0.01 of 1536 px at 0.7 (0.53). The rule selects the latter as closer to defaults.

The local Esri confidence sweep: ≥0.5: 0.43; ≥0.6: 0.46; ≥0.7: 0.49; ≥0.8: 0.51; ≥0.9: 0.53; ≥0.95: 0.48.

Reproducing the area policy

`nbf evaluate --min-area-m2 20` removes predictions and reference polygons below 20 m² after AOI handling and before matching. This is the historical tuning protocol's policy.

`--size-bins-m2 20 50 100` alone summarizes a global matching. Each object's own area determines its class; matched pairs may cross a bin boundary. Threshold summaries therefore need not equal a fresh filtered evaluation.

Keep the ≥20 m² and all-size runs separate, record the area filter and compare only reports with the same policy. The audit reproduces all six city/test ≥20 m² rows by retaining global matched pairs only when both objects meet 20 m². A before-matching filter rerun in the small test AOI gives the same counts. This does not prove historical operation order; the city and 176 tuning variants were not rerun with that filter.

Evidence: tuning protocol and scoring audit.

Frozen settings and held-out inventory scores

A seeded random rule selected a 700 m test square. The frozen settings match the city-wide runs; the area was scored once against its inventory outlines.

Method	Frozen setting	Old Town F1
Esri, city run	as published; regularization off	0.61
SAM 3	building; 1536 px; confidence ≥0.7; regularization off	0.53
Esri, local raw run	confidence ≥0.9; regularization off	0.53

Historical scores reported for at least 20 m²

Method	TP	FP	FN	Precision	Recall	F1	Median IoU	F1 all sizes
Esri, city run	445	16	26	0.97	0.94	0.95	0.76	0.83
SAM 3	430	92	41	0.82	0.91	0.87	0.76	0.76
Esri, local raw run	422	30	49	0.93	0.90	0.91	0.79	0.80

The ≥20 m² columns retain historical published summaries; see the scoring audit for matching policy. The last column covers all sizes. Agreement exceeds Old Town, but its cause is unreviewed. At most 2 of the 136 inventory outlines under 20 m² were found.

Runtime: laptop RTX 4050, 6 GB

Run	Area or tiles	Time
SAM 3, 1024 px	90 tiles, Old Town	105 s; 0.82 s per tile
SAM 3, 1536 px	6,538 tiles, whole city	2 h 41 min; median 1.08 s model time/tile
Esri, local raw run	0.64 km², Old Town	4.9 min
Esri, local raw run	150 chunks, whole city	8 h 46 min model time

SAM 3 used up to 96% of GPU memory, stable through the city-wide run. Earlier extrapolations were about 3.3 h for 1024 px SAM 3 and 15 h for local Esri; the whole-city rows above are measurements.

Methods, provenance and unfinished review

Imagery	Nearmap mosaic, May 2026, 10.16 cm, 21.1 gigapixels, clipped to city limits. The GeoTIFF declared a local CRS in metres; EPSG:2276 in US survey feet was set through a VRT without resampling and checked against the city boundary.
Reference	City of Lewisville inventory: 38,924 source features, primarily dated 2015 or earlier, 33 undated. City-wide evaluation: 38,882 outlines. The reported alignment of about 0.4 m does not establish completeness or temporal accuracy.
SAM 3	Text prompt through SamGeo 1.4.2 at a pinned commit. Tuning used 1024-2048 px tiles; city-wide inference used 1536 px with 192 px overlap, prompt "building", confidence threshold 0.7.
Esri, city run	Building Footprint Extraction - USA (Mask R-CNN), ExtractFeaturesUsingAIModels defaults, including Esri post-processing. Published by the city.
Esri, local raw	DetectObjectsUsingDeepLearning without NMS, retaining confidence scores. The whole city used 150 overlapping chunks of about 1.1 km, retaining outlines centred in each chunk's core.
Scoring	EPSG:26914; shared cleanup with 4 m² minimum; one-to-one matching at IoU ≥0.5; AOI-crossing objects excluded. The protocol requires filtering both sets to at least 20 m² before matching; historical execution requires separate evidence.

Required before an accuracy result

1. Review both complete AOIs against the 2026 imagery without viewing predictions, agreement layers or candidate-selected locations. Record coverage, uncertain areas and the actual reviewer/date.
2. Freeze the reviewed labels and receipt. A receipt passed with `--review-receipt` must identify the same reference and AOI and attest complete, independent review. The software checks the receipt; it cannot perform the review.
3. Score the frozen settings once, with separate all-size and ≥20 m² runs. Do not re-tune using the held-out results.

Status: no completed blind review or receipt is recorded. No reviewed 2026 accuracy result is claimed. Fine-tuning remains unrun; small-structure recall and long-building seams remain development work.

Public evidence: reference_provenance.json, scoring_audit.md, citywide_scores.csv, citywide_by_size.csv, test_area_scores.csv, old_town_scores.csv, old_town_by_size.csv, tuning_final.csv, tuning_protocol.md.

Revised 15 October 2026 from the existing public report, its unchanged screenshots and committed result tables. The original private page was unavailable. This local reconstruction is not a new extraction run, label review or independently verified change inventory.